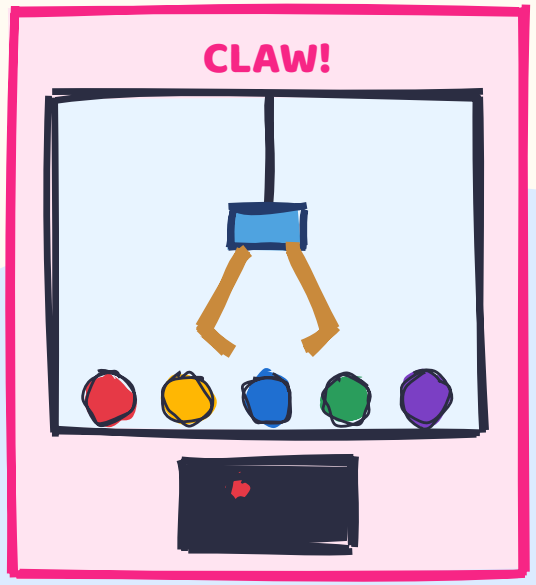
Junior Makers · Class 27

Arcade Claw Machine

Build the robot, then build the game

Servos | Joystick | Mechanisms | Game design



Did you know?
The first claw machines appeared in the 1920s and gave out sweets. Today's are robots with tiny computers inside.

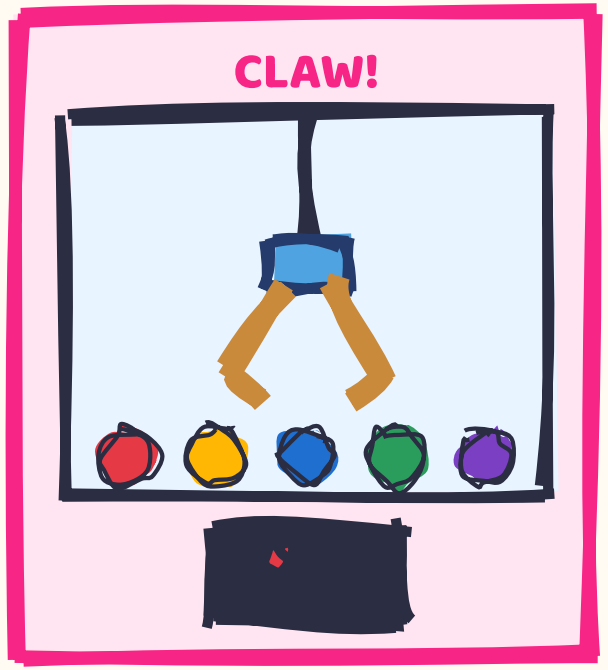
LEARN

Let's think...

Why is the claw so hard to win?

So close!

You line up the claw perfectly, it grabs the toy... and it slips out. Again! Is it your fault, or is the machine **programmed** that way? Today you build your own claw machine and decide the rules.



Bad luck?

Bad aim?

It's programmed!

Today: you build a **claw machine** with two servos and a joystick, then design a **fair** game around it.

Learning objectives

By the end of this class I can...

1	**Servo angles:** move a servo to exact **angles** from 0° to 180°	**Proof:** my servo points to 0°, 90° and 180° on command
2	**Joystick control:** **map** the joystick to the arm's angle for smooth control	**Proof:** pushing the stick swings the arm left and right
3	**Claw mechanism:** build a **pincer claw** from sticks that a servo opens and closes	**Proof:** my claw picks up a paper ball
4	**Game design:** add a **timer and score** and decide what makes the game fair	**Proof:** our game has a 30 s timer and a fair grip

Objective 4 · Game design · video

Video time!



It's not you. Claw machines are rigged.

Vox · about 4 minutes · youtu.be/KnMKCHqXLow

Watch for...

1. What decides how **strong** the claw grips?
2. Who sets those settings: the player or the owner?
3. Is that **fair**? Why or why not?

Scan to watch

After the video: When we build OUR claw machine, should we make it fair? How would you decide how strong the grip is?

Objective 1 · Servo angles

A servo turns to an exact angle

The idea

A **servo** is a motor with a tiny brain inside. You don't tell it 'spin', you tell it an **angle**: 0°, 90°, 180°, and it goes there and **holds**.

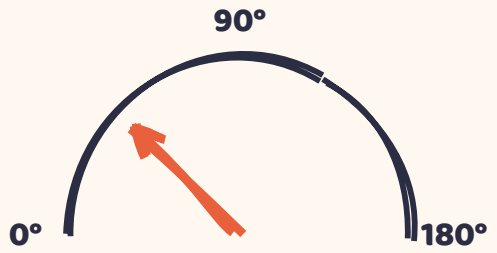
We use **two servos**:

- **Arm servo**: swings the claw left and right
- **Claw servo**: opens and closes the pincers

Strength

SG90 servos are small. They can lift a paper ball, not a can of soda!

Example: a clock hand



If you say 'point to 3 o'clock', a clock hand goes there and stays. A servo is a motor that understands 'point to 90°'. Ordinary motors just keep spinning.

Objective 2 · Joystick control

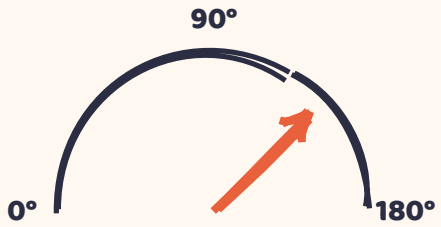
Map the joystick to the angle

The idea

The joystick gives **0 to 1023**. The servo wants **0° to 180°**.
The **map** block converts:
angle = map(joystick, 0…1023, 0…180)

Stick left → 0°, middle → 90°, right → 180°. The arm follows your thumb!

Example: a dimmer switch



joystick 768 → 135°

Turning a dimmer knob a little makes the light a little brighter. The knob's position is 'mapped' to the brightness. Our joystick position is mapped to the arm's angle.

Objective 3 · Claw mechanism

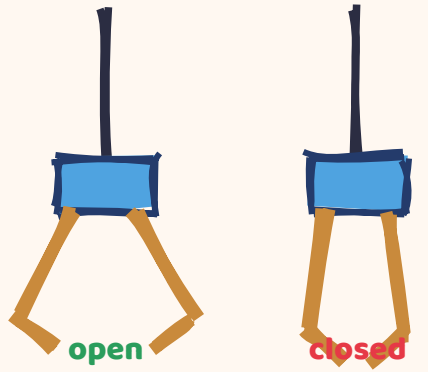
How a pincer claw works

The idea

Two sticks cross like **scissors** at a pivot (a pin or a short screw). Pull the top ends apart and the bottom ends **close**.

The claw servo's arm pulls a short stick link or a string to open and close the pincers. Small movement at the servo = big grab at the tips. This is a **lever**.

Example: kitchen tongs



Squeeze tongs near the hinge and the tips move a lot. That's why tongs can grab a samosa without your fingers getting greasy! Our claw is a robot pair of tongs.

Objective 4 · Game design

Design a fair game

The idea

Now it's a **game**: add rules with code.

- **Timer**: 30 seconds per turn. A countdown on… the buzzer every 10 s!
- **Win**: drop a prize in the chute → LED + victory tune.
- **Grip**: strong enough to really win sometimes.

Real machines can be set to grip weakly most of the time. Should ours?

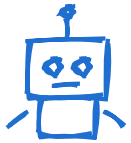
Example: rules of a school sports day



A race is only fun if everyone has a real chance. If the finish line moved away from some runners, nobody would want to play. Games need fair rules that players can see.

Real world

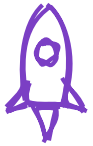
Claws and grippers at work



Factory robots
Pick parts from bins all day



Warehouses
Robots grab boxes for online orders



Space
Robot arms on the space station catch cargo ships



Surgery
Tiny robot grippers help surgeons

? Which is harder for a robot: picking up a box or picking up an egg? Why?

Soft grippers
Some new grippers are soft and squishy, like fingers, so they don't crush delicate things.

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Human claw
2. Our claw kit (per group)
3. Wire and test the servos
4. Steer the arm
5. Build the claw and arm
6. Press to grab
7. Test: is it fun AND fair?

Class game

Human claw

How to play

1. One student is the **claw**, eyes closed, holding two rulers like pincers.
2. A partner is the **joystick**: they can only say **LEFT**, **RIGHT**, **DOWN**, **GRAB**, **UP**.
3. Try to pick up a scrunched paper ball from a desk in 30 seconds.
4. Swap roles. Then try with 'GRAB' only being gentle (weak grip).

Then discuss

What made it hard? How did the weak grip change the game?



Big idea

Control + timing + grip strength decide whether a claw game is fun or frustrating.

Build it

Our claw kit (per group)

Arduino Uno
The machine's brain

2 × SG90 servo
Arm and claw

Joystick module
Steer and grab

Ice-cream sticks + card
The arm and pincers

Glue gun
Teacher or careful use!

Buzzer + LED
Win sound and light

Also: breadboard, jumper wires, and 'prizes': scrunched paper balls or pom-poms.

Glue gun safety
The tip is very hot (around 190 °C). Use it on a mat, never touch the nozzle or fresh glue.

Objective 1 · Servo angles · Build step 1 of 4

Wire and test the servos

Part	Arduino
Arm servo signal	pin 9
Claw servo signal	pin 10
Servo red / brown	5V / GND
Joystick VRx	A0
Joystick SW (button)	pin 2
Buzzer / LED	11 / 12

Setup

- servo pin 9 angle 90 (arm)
- servo pin 10 angle 90 (claw)

Loop

- servo pin 9 angle 0 · wait 1 s
- servo pin 9 angle 90 · wait 1 s
- servo pin 9 angle 180 · wait 1 s

Before building!
Set both servos to 90° FIRST, then attach the arms. Now 90° is the middle.

Objective 2 · Joystick control · Build step 2 of 4

Steer the arm

```
Loop
    set angle to map( analog read A0 , 0, 1023, 20, 160 )
    servo pin 9 angle angle
    wait 20 ms
```

Why 20 to 160, not 0 to 180?

The arm might hit the walls of the cabinet at the ends. Limiting the range protects your machine. Engineers call these **soft limits**.

Jittery?

Servos twitch if the number keeps changing by 1. Only move when the angle changes by more than 2.

Objective 3 · Build step 3 of 4

Build the claw and arm

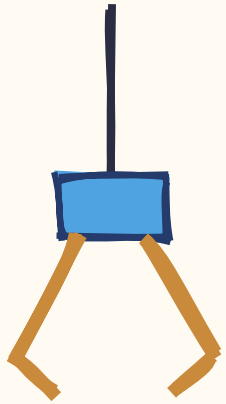
a Glue 3 ice-cream sticks end to end for the arm. Glue the arm to the arm servo's horn.

b Make 2 pincers from sticks, crossed at a pivot (split pin or toothpick through holes).

c Mount the claw servo at the end of the arm. Connect its horn to the pincers with a short stick link.

d Test by hand: does turning the claw servo open and close the tips?

e Build a card cabinet round it, with a prize area and a 'win' chute.



Check: Turning the claw servo by about 40° opens and closes the pincers fully.

Let glue cool for 1 minute before moving parts. Hot glue joints are weak while warm.

Objective 3 · Claw mechanism · Build step 4 of 4

Press to grab

Setup: set clawOpen to true · servo pin 10 angle 120

Loop

(steer arm as before)

if digital read pin 2 = LOW (stick pressed) then

if clawOpen → servo pin 10 angle 70 (close) · set clawOpen false

else → servo pin 10 angle 120 (open) · set clawOpen true

wait 300 ms (so one press = one change)

Toggle

Each press flips the claw: open → closed → open. The variable **clawOpen** remembers which.

Tune the grip

Close angle 70 too tight? The servo buzzes. Try 75 or 80.

Objective 4 · Game design · playtest

Test: is it fun AND fair?

Playtest card	Result
Turns played	
Prizes won	
Win rate (wins ÷ turns)	
Did players feel cheated?	
One change we made	

What's a fair win rate?

Discuss: 1 win in 2 turns? 1 in 10? Should players be told the win rate before they play?

Honesty

If we sold turns for money, hiding a weak grip would be unfair. Good designers are open about the rules.

Test like an engineer

Bug hunt

Bug	Likely cause	Did it happen?
Servo shakes or buzzes	angle pushing against something	
Arduino resets when servos move	two servos draw too much from USB	
Arm moves the wrong way	swap 20 and 160 in map	
Claw toggles twice per press	wait after press too short	
Pincers won't close fully	link too short / pivot too tight	

Debug checklist

Unplug the servos and test the joystick numbers alone first.

If the board resets, power the servos from a 4×AA pack (joined GND) or move only one servo at a time.

Level up!

Claw challenges

Choose your challenge

- **Up and down:** add a third servo or a string winch to lower the claw.
- **Countdown:** an LED that blinks faster as time runs out.
- **High score:** count wins across the class and show the leader.
- **Accessible:** a big-button version for younger students.

Think like an engineer

Why do real claw machines have the claw swing a little when it lifts?

Boss level

Make a 'demo mode' that plays by itself when nobody is using it.

Objective check

Can I really do it? Prove it.

01	Make your arm servo point to 0°, 90° and 180°.	○ ○ ○
02	Explain your map block. Why 20 to 160?	○ ○ ○
03	Show your claw picking up a prize. What lever makes it work?	○ ○ ○
04	Tell us your timer, your win rate, and whether your game is fair.	○ ○ ○

Rate yourself

Colour one circle each. Red: not yet. Amber: with help. Green: I can explain it to a friend.

Teacher: Claw arcade: every team plays 3 other machines and votes for the most fun AND fair.

Show what you know

What did we learn?

- [x] Servos turn to an exact angle and hold it
- [x] map converts joystick numbers into angles
- [x] Soft limits protect the machine
- [x] Pincers are levers: small turn, big grab
- [x] Fair games have rules players can trust

Exit ticket: why do we set servos to 90° before attaching the arms?

At home

Look at tongs, scissors and pliers at home. Where is the pivot? Draw one.



Next class: Wind Turbine Power Lab!